Shanto-Mariam University of Creative Technology

Project Title: RecipeHub-Complete Flutter Recipe App
Course Name: Mobile Application & Development
Course Code: 4131

Software Requirement Specification(SRS)

Submitted By	Submitted To
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1. Introduction

1.1 Purpose

The purpose of the RecipeHub mobile application is to provide a user-friendly platform where users can discover, save, and manage recipes efficiently.

This system helps users easily search recipes, learn cooking, and organize their favorite dishes in one place.

1.2 Scope

RecipeHub is a Flutter-based mobile application that allows users to:

- Browse recipes by category
- Search recipes using ingredients or name
- View recipe details (ingredients + instructions)
- Save favorite recipes
- Upload their own recipes
- Rate and review recipes
- Admin manage users and recipes

1.3 Objectives

- Develop a responsive mobile application
- Provide an easy cooking learning platform
- Implement authentication system
- Enable user-generated content (upload recipes)
- Ensure smooth user experience

1.4 Target Users

Home Cooks

- Individuals who want to cook at home
- Users interested in learning new recipes

Food Lovers

- Users who enjoy exploring different dishes

Admin

- Responsible for managing system data and users

1.5 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
CRUD	Create, Read, Update, Delete

2. Overall Description

2.1 Product Perspective

RecipeHub is a standalone mobile application developed using Flutter.

- **Frontend:** Flutter
- **Backend:** Firebase / Supabase
- **Database:** Firestore / PostgreSQL

2.2 Product Functions

The main functionalities of the system include:

- User registration and login
- Browsing recipes
- Searching recipes
- Viewing recipe details
- Saving favorite recipes
- Uploading new recipes
- Rating and reviewing recipes
- Admin management system

2.3 User Classes and Characteristics

User

- Registers and logs in

- Browses and searches recipes
- Saves and uploads recipes
- Provides reviews and ratings

Admin

- Manages recipes
- Removes inappropriate content
- Monitors user activities

2.4 Operating Environment

- Platform: Android / iOS
- Framework: Flutter
- Backend: Firebase / Supabase
- Internet: Required

2.5 Design and Implementation Constraints

- The application must be developed using Flutter
- Backend must use Firebase or Supabase
- The application must support Android devices
- Development time is limited (academic project)
- Internet connection is required

2.6 Assumptions and Dependencies

- Users must have a smartphone with internet access
- Backend services must remain operational
- Users must provide valid credentials during registration

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement	Priority
FR1	User registration	High
FR2	User login	High
FR3	Browse recipes	High
FR4	Search recipes	High

FR5	View recipe details	High
FR6	Save favorite recipes	Medium
FR7	Upload recipes	Medium
FR8	Rate and review recipes	Medium
FR9	Admin management	High

3.2 Non-Functional Requirements

ID	Requirement
NFR1	App should load within 3 seconds
NFR2	Secure authentication system
NFR3	User-friendly interface
NFR4	Support multiple users
NFR5	Ensure data consistency

3.3 External Interface Requirements

User Interface

- Splash Screen
- Login & Registration Screen
- Home Screen
- Recipe Category Screen
- Recipe Detail Screen
- Favorite Screen
- Upload Recipe Screen

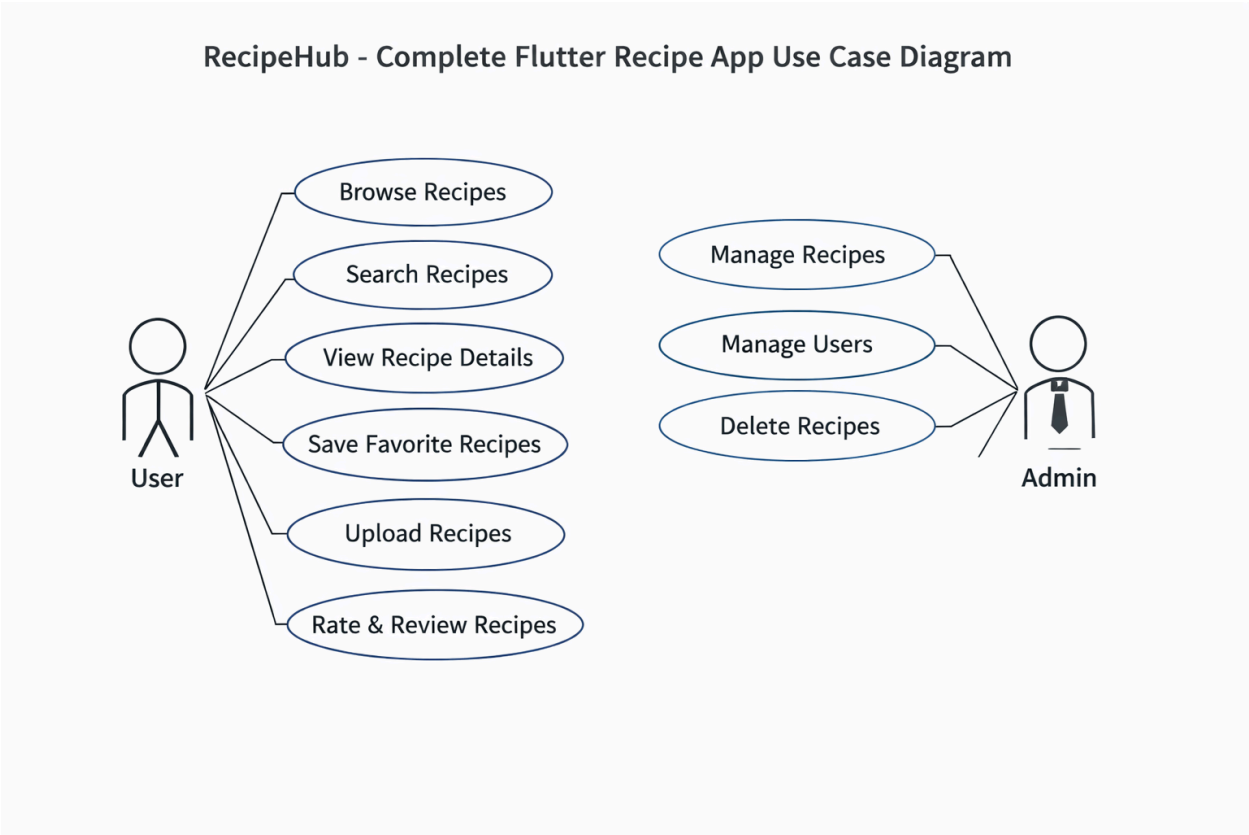
Software Interface

- Firebase / Supabase API
- Internet connection

Hardware Interface

- Android / iOS devices
- Internet connectivity

4. System Model



Use Case Overview

Actors:

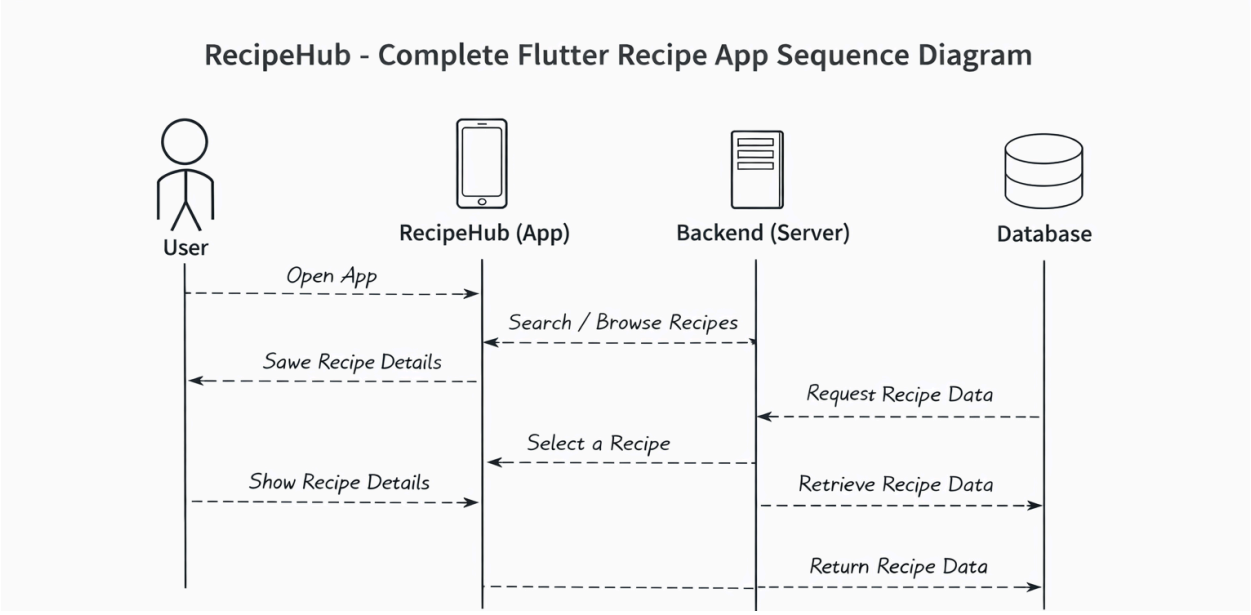
- User
- Admin

User Actions:

- Browse recipes
- Search recipes
- View recipe details
- Save favorite recipes
- Upload recipes
- Review and rate recipes

Admin Actions:

- Manage recipes
- Monitor users
- Delete inappropriate content



Steps:

1. User opens the app

2.User searches. Or browses recipes

3.User selects a recipe

4.RecipeHub sends request to backend

5.Backend fetches data from database

System Flow (Corrected)

1. User opens the application
2. User browses or searches for recipes
3. User selects a recipe
4. The app sends a request to the backend
5. The backend retrieves data from the database
6. The system sends the response to the app
7. The app displays the recipe details
8. User can save or review the recipe

5. Project Management

5.1 Development Tools

- Flutter
- Firebase / Supabase
- Android Studio / VS Code
- GitHub

- **5.2 Timeline**

Team Members	**Riya**	**Roman**
Week 1	Planning & Requirement Analysis	Planning & Requirement Analysis
Week 2	UI Design	UI Design
Week 3	Home screen, login/signup UI design	Recipe list screen, detail screen UI design
Week 4	UI coding (login, signup, home)	UI coding (recipe details, category, search UI)
Week 5	Signup & login system implement	User validation, error handling, logout system
Week 6	UI + backend connect (auth screens)	Security rules, session management
Week 7	Add recipe feature, image upload	View recipe, recipe list, category filter
Week 8	Search functionality, favorite option	API/database integration, data fetching
Week 9	UI testing, bug finding	Backend testing, performance optimization
Week 10	UI bug fix, responsiveness fix	Logic bug fix, database fix
Week 11	Final UI polish, documentation	Final testing, deployment preparation

6. Appendix

- Flutter Documentation
- Supabase Documentation
- Course Materials
- Online Resources